500 Labs

Antigravity Workshop & POS Master Guide

Vehicle Repair, Collision Estimation & Spare Parts POS System

Unified Project Brief + Agent Instructions + Phase-by-Phase Implementation Guide

React + Vite + TypeScript | Tailwind CSS | shadcn/ui | NestJS | PostgreSQL

Prepared for implementation by an Antigravity coding agent

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1. Purpose

This document combines the two requested artifacts into one plain, AI-readable guide: the Vehicle Repair & Spare Parts POS project/agent instructions and the phase-by-phase implementation plan. It is intended to be given to an Antigravity coding agent and followed as the strict project operating contract.

The system defines a lean, high-speed web application for vehicle repair workshops covering on-the-fly manual Customer & Vehicle entry, Quotations, Accident Repair Estimates, Workshop Job Cards, Final Invoices, Counter/Showroom POS (for direct walk-in spare parts, lubricants, and quick-service billing), a Parts & Labor Preset Catalog, embedded Sales/Cost/Gross Profit tracking, payments, WhatsApp/A4 print exports, reporting, and audit logs.

2. Source Baseline

Use the uploaded workshop document templates (*Accident Repair Quotation* and *Accident Repair Estimate*) and explicit project decisions as the authoritative business-requirements baseline. Do not silently replace, remove, or invent complex modules that were excluded.

- **No Mandatory CRM Prerequisite:** Users must be able to type Customer Information, Vehicle Details, and Insurance Company names directly onto the sheet without pre-creating customer accounts.
- **3-Section Repair Sheet Structure:** Every Quotation, Estimate, Job Card, and Invoice must preserve the exact three line-item groups: *Labor - Remove & Reassemble*, *Labor - Repair*, and *Replacement* (Spare Parts).
- **Embedded Financial Engine (Not a Separate Accounting Module):** Sales Revenue, Buying/Direct Cost, Gross Profit, Margin %, Tax, Discounts, Advances, and Balances are calculated directly on the repair documents and POS transactions.
- **Integrated Counter POS:** Walk-in customers buying spare parts, oils, batteries, or quick counter services are billed through a fast POS screen that shares the same Parts & Labor Catalog and feeds into the unified Daily Sales, Cost & Gross Profit dashboard.
- **Exclusions:** Full double-entry general ledger (COA/Trial Balance), complex multi-warehouse bin management, HR/Payroll, and mandatory customer registration portals are strictly excluded.

3. Non-Negotiable Agent Rules

- Never build the entire Workshop & POS system in one task. Work strictly phase-by-phase (Phase 0 through Phase 10).
- First inspect the repository, `package.json`, routes, components, styles, dependencies, state, and existing mock data.
- Create a requirements matrix and trace every feature to the workshop templates or explicit project decisions.
- The **Parts & Labor Catalog Master** is the single source of truth for preset items and barcodes, used by both Workshop Sheets and Counter POS. However, Workshop Sheets must always permit 100% freeform manual typing of descriptions and prices for custom accident repairs.
- Do not force Customer Master registration. Store typed customer and vehicle fields directly on the job/POS record, while offering optional non-blocking auto-complete from previous jobs by Vehicle Number or Phone.
- Use typed repository/service abstractions. Mock implementations are allowed during frontend development; the architecture must be ready for NestJS/PostgreSQL.
- Do not put business data or profit math directly inside page components.
- Every action needs loading, success, error, and disabled states.
- Use toast feedback for non-blocking events and confirmation dialogs for destructive or stage-conversion actions.
- Centralize permission checks (especially hiding Buying Cost and Gross Profit from standard cashiers/mechanics and customer printouts).
- Preserve historical transaction snapshots (description, selling price, buying cost, tax, discount) on every repair line and POS cart line so catalog edits never alter past invoices.
- Keep management and estimation screens desktop-friendly, and make Job Card / Estimator views tablet- and mobile-friendly.
- Do not move to the next phase until the current phase is implemented, tested, reviewed, and documented.

4. .agent Directory

- `.agent/AGENT.md` - master operating instructions
- `.agent/RULES.md` - engineering, UX, print-privacy, and business-rule constraints
- `.agent/WORKFLOW.md` - phase workflow and completion gate
- `.agent/ARCHITECTURE.md` - frontend and NestJS integration architecture
- `.agent/REQUIREMENTS_MATRIX.md` - template and MVP feature traceability
- `.agent/DOMAIN_MODEL.md` - entities, relationships, embedded profit math, and state transitions
- `.agent/UI_GUIDELINES.md` - design system, A4 print layout, thermal receipt, and responsive rules
- `.agent/TESTING_STRATEGY.md` - QA strategy
- `.agent/INTEGRATION_CONTRACT.md` - API-ready contracts
- `.agent/skills/` - reusable domain/engineering skills

- `.agent/agents/` - specialized sub-agent definitions

Recommended sub-agents: Orchestrator/Project Manager, Frontend Architect, Catalog & Presets, Workshop Document Workflow, Job Card & Floor Operations, POS & Counter Sales, Embedded Finance & Gross Profit, Print/Thermal & WhatsApp Export, Reporting, Security/Permissions, API Integration-Readiness, QA/Test, and UI/UX Enhancement.

5. Architecture

Frontend

- React + Vite + TypeScript.
- Tailwind CSS + shadcn/ui.
- Feature/domain-oriented folders (`workshop`, `pos`, `catalog`, `reports`, `auth`).
- React Hook Form + Zod for document headers, line arrays, and payment dialogs.
- Centralized routing, permissions, repository contracts, error handling, and notifications.
- Reusable `VehicleLookupInput`, `CatalogItemCombobox`, `RepairLineTable`, `PosCartTable`, `MarginSummaryBar`, `A4PrintLayout`, and `ThermalReceiptLayout` components.

Integration-Ready Data Layer

- Define `CatalogRepository`, `WorkshopJobRepository`, `PosSaleRepository`, `PaymentRepository`, and `ProfitReportRepository` interfaces.
- Use stateful mock repositories now and API-backed repositories later.
- Keep DTO/domain types explicit and compatible with NestJS contracts.
- Do not create a competing frontend database/source of truth.
- Represent pagination, filtering, vehicle plate lookup, validation errors, and authorization errors in the repository contract.

6. Parts & Labor Catalog Master (Shared by Workshop & POS)

To support fast counter POS checkout and quick workshop estimation without forcing complex inventory barriers, the system maintains a unified Parts & Labor Catalog.

- Core fields: `Item ID`, `SKU / Part Number`, `Barcode`, `Item Type` (`SPAREPART`, `LUBRICANT`, `LABOR_REMOVE_REASSEMBLE`, `LABOR_REPAIR`), `Name / Description` (required), `Category`, `Brand / Vehicle Compatibility`, `UOM`, `Buying Cost (Rs.)`, `Selling Price (Rs.)`, `Default Discount %`, `Available Stock Qty` (optional tracking for POS items), `Low Stock Alert Level`, and `Active Status`.
- Routes: `/catalog`, `/catalog/new`, `/catalog/:id`, `/catalog/:id/edit`.
- List: search by part name, vehicle model compatibility, SKU, or barcode scan.
- In Workshop Sheets: users can select from this catalog to auto-fill description, selling price, and hidden buying cost, OR type a completely custom description and price on the fly.

- In Counter POS: users scan barcodes or search parts/lubricants to add items to the POS cart with automatic Selling Price and Buying Cost capture.

7. On-the-Fly Vehicle & Customer Header (Zero-CRM Friction)

Per explicit project rules, there is no mandatory Customer Management module blocking operations. Customer and vehicle data are entered directly onto the document header.

- **Workshop Sheet Header Fields (Exact Template Match):**
  - Row 1: `Date` | `Job Number` (auto-generated sequential, e.g., `JOB-2026-0001`)
  - Row 2: `Vehicle Number` (e.g., `CBA-1234`) | `Nature of Repair` (Accident, Mechanical, Electrical, Scanning, Tinkering, Auto Paint)
  - Row 3: `Vehicle Make` | `Chassis Number`
  - Row 4: `Vehicle Model` | `No of Days Completion`
  - Row 5: `Year of Make` | `Vehicle Color`
  - Row 6 (Full Width): `Name of Insurance`
  - Row 7 (Full Width): `Customer Information` (Name, Phone Number, Address typed directly)
- **Smart Vehicle Plate Auto-Fill:** Typing a previously seen `Vehicle Number` offers a one-click suggestion to auto-populate Make, Model, Year, Color, Chassis Number, and Customer Information from the last job history without requiring a separate CRM screen.
- **POS Walk-in Customer Fields:** Optional single-line input for `Customer Name / Phone` and `Vehicle No` on the POS screen (defaults to `Walk-in Counter Sale`).

8. Roles

Role	Primary Responsibility
Owner / Director	Full visibility into Total Sales, Buying Costs, Gross Profit, Margin %, overrides, and all workshop/POS operations.
Workshop Manager	Creates/approves Quotations & Accident Estimates, manages Job Cards, enters part buying costs, converts to Invoice, views daily sales and gross profit.
Service Advisor / Estimator	Intakes vehicles, types Quotations and Estimates, converts to Job Cards, prints/shares documents via WhatsApp.
Cashier (POS & Billing)	Operates Counter POS for spare parts/lubricants, collects Workshop Invoice payments, manages POS cash drawer/shifts, prints thermal receipts and A4 invoices.
Technician / Floor Lead	Views Workshop Job Cards (with all prices hidden), updates repair task completion checkboxes, and flags waiting-for-parts status.

9. Core Business Workflows

Workshop Document Workflow (Quotation → Estimate → Job Card → Invoice)

- Create a new sheet in **Quotation** (for general/customer inquiries) or **Estimate** (for *Accident Repair Estimate* submitted to insurance assessors).
- Enter vehicle/customer header fields manually and add rows under *Labor - Remove & Reassemble*, *Labor - Repair*, and *Replacement*.
- Toggle **Admin Cost & Profit Mode** to record the actual `Buying Cost (Rs.)` of replacement parts and direct technician/sublet costs per row. The system calculates live `Gross Profit (Rs.)` and `Gross Margin %` on screen.
- Print A4 Quotation/Estimate or click **Share via WhatsApp** (opens WhatsApp share link with formatted summary and PDF download prompt; no paid WhatsApp API required).
- Once approved by the customer or insurance assessor, click **Convert to Job Card**. The system preserves an immutable snapshot of the original estimate, switches the document title to *Workshop Job Card*, and hides all monetary figures on the printed floor sheet.
- During repair, supplementary parts or labor discovered after stripping the vehicle can be appended to the sheet.
- Upon completion, click **Convert to Invoice**. Record any Discount, Tax, Advance Paid, and Insurance vs. Customer Excess split billing. Collect payment and issue an A4 Invoice or thermal payment receipt.

Counter / Showroom POS Workflow

- Cashier opens the **POS Screen (`/pos`)** for walk-in spare part sales, engine oil/lubricant purchases, batteries, or quick fixed-price counter services (e.g., Quick Battery Check, Wiper Replacement).
- Scan barcode or search the Parts & Labor Catalog; support adding an on-the-fly custom POS item if not in the catalog.
- Adjust quantity or line discount within permitted limits. Buying cost is attached automatically behind the scenes from the catalog (or typed inline) to ensure accurate Gross Profit calculation.
- Select payment method (`Cash`, `Card`, `Bank Transfer / QR`, or `Split Payment`), enter tendered cash, calculate change, and complete sale.
- Issue an 80mm thermal receipt (or A4 bill) and automatically log the sale's revenue, cost, and gross profit into the daily financial summary.
- Support POS Shift Open/Close (opening float, cash in/out for minor garage expenses/petty cash, and end-of-day cash reconciliation).

Embedded Sales, Cost & Gross Profit Workflow

- Every completed Workshop Invoice and POS Sale computes:
  - `Gross Sales` = Sum of Labor Selling Price + Sum of Parts Selling Price
  - `Net Sales Revenue` = Gross Sales − Discounts
  - `Total Direct Cost` = Sum of Replacement Part Buying Costs + Sum of Direct Labor/Sublet Costs
  - `Gross Profit` = Net Sales Revenue − Total Direct Cost
  - `Net Shop Profit` = Gross Profit − Daily Garage Expenses (recorded via simple expense log / POS cash-out)
- Crucial Privacy Rule: Buying Costs and Gross Profit are strictly for Owner/Manager visibility and must never appear on Customer Quotations, Insurance Estimates, Job Cards, Invoices, or POS Receipts.

10. Document Lifecycle & Job Status State Machine

Do not implement document stages or workshop statuses as scattered booleans. Use explicit state machines:

- **Document Stage (`docStage`):** `QUOTATION` → `ESTIMATE` → `JOB_CARD` → `INVOICE` (plus `CANCELLED`).
- **Operational Floor Status (`jobStatus`):** `DRAFT` → `PENDING_INSURANCE_APPROVAL` → `APPROVED_IN_PROGRESS` → `WAITING_FOR_PARTS` → `QUALITY_CHECK` → `READY_FOR_DELIVERY` → `COMPLETED_DELIVERED`.
- **Payment Status (`paymentStatus`):** `UNPAID` → `PARTIALLY_PAID` (Advance/Insurance Pending) → `PAID` → `REFUNDED_VOID`.

Every stage transition, discount override, or invoice reopening must create an audit record containing actor, action, timestamp, previous state, new state, and comment.

11. Phase-by-Phase Implementation

Phase 0 - Analysis & Planning

Inspect repository and uploaded templates. Create `.agent/` files, architecture, domain model, requirements matrix, and open questions. Do not start large feature work.

Phase 1 - Design System & Print Abstractions

Set Tailwind/shadcn conventions, typography, high-contrast workshop tables, keyboard-navigable line inputs, dialogs, toast notifications, A4 print wrapper (`@media print`), and 80mm thermal receipt wrapper.

Phase 2 - App Shell & Role Permissions

Routing, login abstraction, protected routes, role-based permission guard (controlling cost/profit visibility), sidebar, header, quick action buttons (`+ New Repair Sheet`, `Open POS`), and role dashboards.

Phase 3 - Parts & Labor Catalog + Vehicle History Lookup

Build the unified Parts & Labor Catalog (`/catalog`) with SKU/barcode support, Selling Price, and Buying Cost. Build the non-blocking Vehicle Number lookup helper that auto-fills vehicle specs from past jobs while keeping all fields 100% manually editable.

Phase 4 - Workshop Quotations & Accident Repair Estimates

Implement the unified Repair Sheet (`/workshop/sheet`) with the exact 12-field vehicle/insurance/customer header and the 3 dynamic tables (*Labor - Remove & Reassemble*, *Labor - Repair*, *Replacement*). Add live subtotals, Admin Buying Cost & Gross Profit columns, A4 print layout matching the uploaded templates (including the top service banner and *"Computer generated estimate and does not carry a signature"* footer), and WhatsApp share navigation.

Phase 5 - Job Card Conversion & Workshop Floor Board

Implement one-click conversion from Quotation/Estimate to **Job Card**, saving an immutable snapshot of original estimated amounts. Build the price-redacted Job Card print view (showing technician assignment, target days, and task checkboxes) and the Workshop Status Board (`/workshop/board`) filtering jobs by floor status.

Phase 6 - Workshop Invoicing, Split Billing & Collections

Implement one-click conversion from Job Card/Estimate to **Invoice**. Support supplementary line items, Discounts, Taxes, Advance Payments, Insurance Approved Payable vs. Customer Excess split billing, Balance Due tracking, payment collection, and invoice printouts.

Phase 7 - Counter / Showroom POS & Shift Cash Management

Build the dedicated Counter POS interface (`/pos`) for direct walk-in spare parts, lubricants, and quick services. Include barcode/catalog search, custom item entry, cart calculations, hidden buying-cost capture for profit tracking, cash/card/transfer tender modal, 80mm thermal receipt print, POS Shift open/close, and petty-cash expense payouts.

Phase 8 - Embedded Finance, Sales, Cost & Gross Profit Center

Build the unified Financial & Gross Profit module (`/finance/profit-ledger`) combining Workshop Invoices and POS Sales. Include a simple Daily Shop Expense entry screen (rent, electricity, consumables, sublet bills) and real-time calculation of Gross Sales, Parts Cost, Labor Revenue, Gross Profit, Margin %, and Pending Insurance/Customer Receivables.

Phase 9 - Reports & Analytics Dashboards

Implement Date-Range Sales & Gross Profit Report (broken down by Workshop Labor, Workshop Replacements, and Counter POS), Pending Insurance Claim Aging Report, Technician Turnaround Report, and Fast-Moving Parts / Low-Stock Alert Report.

Phase 10 - Full QA, Print Audit & Integration Readiness

End-to-end workflow testing, cost/margin privacy leak audit on all print templates, keyboard navigation check, responsive review, regression testing, and NestJS/PostgreSQL API contract verification.

12. Testing and QA

- Testing is a separate sub-agent responsibility.
- Test manual entry of a complete Accident Repair Estimate with zero pre-existing customer or catalog records.
- Test state transitions: `Quotation → Estimate → Job Card → Invoice` verifying zero data loss across all three tables.
- Test the **Cost Privacy Invariant**: verify that `Buying Cost` and `Gross Profit` never render in the DOM print preview or PDF export for Quotations, Estimates, Job Cards, Invoices, or POS Receipts.
- Test Gross Profit math across Workshop Invoices and Counter POS sales: verify `Net Sales - Total Buying/Direct Cost = Gross Profit` to the cent.
- Test Insurance Split Billing math: verify `Grand Total - Advance Paid - Insurance Approved Amount = Customer Balance Due`.
- Test Counter POS barcode scanning, cart updates, shift cash-in/cash-out, and thermal receipt abstraction.
- Test keyboard accessibility: pressing `Enter` in the last line-item amount input must append a new numbered row cleanly.

13. UI Enhancement Agent

- UI enhancement is separate from feature implementation.
- Improve visual hierarchy, spacing, table readability, loading/error/empty states, and high-speed keyboard ergonomics without changing business rules.
- Ensure the A4 print layout strictly mirrors the uploaded workshop document templates.
- Do not remove required fields/actions or introduce mandatory CRM modals that slow down manual typing.
- Prefer reusable shadcn/Tailwind patterns over one-off designs.

14. Definition of Done

- Requirement traced to workshop templates or recorded project decision.
- Business rule and Gross Profit math implemented in reusable domain/service layer.
- Permissions centralized and enforced (Cost/Margin restricted to Owner/Manager).
- Data access behind typed repository/service boundary.
- Loading, error, empty, and success states exist.
- A4 Print and 80mm Thermal Print views verified with zero UI chrome or cost leakage.

- Relevant unit and workflow tests pass.
- Responsive and keyboard-navigation checks complete.
- Sensitive actions (stage changes, price overrides, invoice voids) are auditable.
- UI enhancement review complete.
- Documentation updated and phase completion report written.
- Next phase starts only after current phase approval.

15. Completion Report Template

- Phase
- Completed Features
- Files Changed
- Business Rules & Financial Formulas Added
- Routes Added
- Components Added
- Services/Repositories Added
- Types/Interfaces Added
- Tests Added and Results
- Print / PDF Privacy Verification Result
- Known Issues & Open Questions
- Integration-Readiness Notes
- Next Phase Prerequisites

16. Master Instruction to Antigravity

Act as the lead engineer and orchestrator for this Vehicle Repair & POS ERP. Follow this guide and the uploaded workshop templates as the project contract. Do not implement the whole system at once. Start with repository analysis and Phase 0. Build the architecture, print abstractions, and shared Parts & Labor Catalog before transaction modules. Preserve extreme speed for workshop staff: allow 100% manual typing of customer, vehicle, insurance, and repair line items directly on the sheet without mandatory CRM registration, while using the shared Catalog for quick-add presets and Counter POS barcode checkout. Track Buying Cost, Selling Price, and Gross Profit directly inside every Workshop Sheet and POS Sale, strictly hiding internal costs and profit margins from customer printouts. Keep the frontend integration-ready for NestJS/PostgreSQL using typed repositories/services and mock implementations during frontend development. Use React/Vite/TypeScript, Tailwind, and shadcn/ui. Use separate sub-agents for testing and UI enhancement. After every phase, test, document, review, and report completion before continuing.

17. Source Notes

This guide is grounded in the uploaded *Vehicle Repair Quotation 1.docx* and *Vehicle Repair Estimate 1(1).docx* templates and explicit MVP architectural decisions. In-scope modules are: Unified Workshop Document Engine (Quotation, Estimate, Job Card, Invoice), On-the-Fly Vehicle/Customer Entry, Shared Parts & Labor Catalog, Counter/Showroom POS, Embedded Sales/Cost/Gross Profit & Expense Tracking, Insurance Split Billing, A4 & Thermal Print Abstractions, and Analytics Dashboards.

The system's two core operational flows are: (1) **Workshop Flow:** Vehicle Intake & Manual Header Entry → Quotation / Accident Estimate → Customer/Insurance Approval → Workshop Job Card → Final Invoice → Split/Full Payment & Gross Profit Posting; and (2) **Counter POS Flow:** Barcode/Part Lookup → POS Cart → Payment Tender → Thermal Receipt & Gross Profit Posting.